Forward-Registered, Auditable LLM-Assisted Research: A Reliability Methodology, with a Capital-Markets Testbed

Yixing Zheng

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Abstract

Large language models are increasingly used as decision engines for open-ended research, but they are unreliable in characteristic ways: they generate plausible but unsupported claims, amplify narrative priors, over-admit candidates, and — when evaluated retrospectively — overfit data that postdates their training. We study this as a problem of **AI reliability and auditability**, using open-ended thematic investment research as a concrete, adversarial testbed. We present **a forward-registered, agent-assisted research framework** that wraps LLM-driven discovery in deterministic gates and a forward pre-registration protocol, so that a hypothesis can be admitted to a forward-evaluation portfolio only after surviving falsifiable, dated-evidence checks and being evaluated *forward* on paper trading rather than by a fragile backtest.

The framework contributes four components.

1. A **framework**: a domain-agnostic bottleneck graph in which nodes are physical chokepoints and adding a domain is mostly a matter of configuration (a new bottleneck map) rather than new modeling code, with a decomposed confidence model (bottleneck × purity × demand × valuation × crowding).

2. A **protocol** for disciplined judgment: deterministic gates, source-tier rules, a bear case written before any recommendation, overlap penalties, and government policy entered as *dated evidence and a reversal trigger, never as a multiplicative factor* — together an explicit mechanism for *rejecting* candidates, not merely surfacing them.

3. **Forward-QPOP**, a **hash-chained**, append-only pre-registration ledger for hypothesis admission, belief updates, and exits — the contribution is the *forward-locked hypothesis contract* (dated evidence + entry/update/exit triggers committed before the evaluation window), not the tamper-evidence alone.

4. A **pre-registered evaluation with pilot evidence** — a low admission rate, a held-out rejection-quality audit, discipline ablations against ungated baselines, and a cross-domain portability probe — released as an open framework, with forward outcomes left to the evaluation window.

Rather than a return claim, our central contribution is methodological: **this validates process discipline, not investment profitability** (which awaits the forward window). We test whether a disciplined, forward-registered agent **rejects most of the candidates it surfaces** *for defensible reasons.* We report *pilot* discipline metrics from early candidate batches — a low admission rate and a **held-out rejection-quality audit** (raw precision 0.93) — which *motivate* the pre-registered

forward evaluation, not confirm it; forward returns are not yet observed. The method makes each admitted position explainable from its content-hashed ledger entry, whereas ungated LLM screeners lack an explicit rejection mechanism. We release the framework, a worked AI-supply-chain example with a real candidate flow, and a template for new domains.

1 Introduction

Large language models have dramatically lowered the cost of thematic investment research. A practitioner can now surface dozens of candidate ideas—bottleneck plays, supply-chain theses, policy-driven opportunities—within minutes of a prompt. The danger, however, is symmetric: the same fluency that makes LLMs productive *sourcing* tools makes them unreliable *decision engines*. LLMs are rewarded, by training and by user expectation, for producing confident, coherent narratives. Left ungated, an LLM screener over-admits by construction—it is optimised to find ideas, not to refuse them—and the narratives it generates are difficult to audit, reproduce, or falsify after the fact.

Existing work on financial LLM agents has concentrated on improving recommendation quality: better prompts, chain-of-thought reasoning, retrieval-augmented generation, or multi-agent debate [8, 28]. These advances are real, but they address the wrong bottleneck. The limiting constraint for a practitioner deploying LLM-assisted research is not the *quality of individual suggestions*; it is the absence of a disciplined, auditable mechanism for *rejecting* the majority of what the model surfaces. Without such a mechanism, every narrative becomes a plausible trade, and the researcher is left arbitrating between fluent confabulations rather than falsifiable hypotheses. What is needed is not better prompts—it is **auditable restraint**.

This paper describes a methods framework that enforces that restraint through five interlocking mechanisms: a domain-portable **bottleneck graph** that decomposes a thematic thesis into physical chokepoints; a **confidence decomposition** that derives a position tier from scored node-and-ticker attributes rather than assertion:

$$\begin{aligned}\text{confidence} = {} & \underbrace{\text{bottleneck\_score}}_{\text{node: how binding}} \times \underbrace{\text{exposure\_purity}}_{\text{ticker: proxy quality}} \times \underbrace{\text{demand\_score}}_{\text{node: order visibility}} \\ & \times \underbrace{\text{valuation\_adjustment}}_{\leq 1} \times \underbrace{\text{crowding\_adjustment}}_{\leq 1}\end{aligned}$$

a **cheap-source → deterministic-gate → expensive-evaluate** funnel that concentrates costly inference on the small fraction of candidates that survive reproducible, LLM-free gate checks; a **bear-case-before-recommendation** evaluation discipline that counteracts sycophancy; and a **forward-QPOP pre-registration ledger** that commits hypotheses, dated evidence, and measurable exit triggers *before* any capital-allocation claim, foreclosing HARKing and backtest overfitting.

The pilot empirical result reported in this draft is deliberately not a return claim. It is the **admission rate**: a disciplined agent *rejects most of what it surfaces*. In our initial pilot batch, fewer than 10% of LLM-surfaced candidates were admitted to the forward ledger; the remainder were rejected via deterministic gates or bear-case evaluation, with an auditor-verified rejection precision of 0.93 on a 14-rejection sample. We report this restraint figure alongside ablation comparisons—ungated screener, debate-only, no-QPOP-lock, no-overlap-penalty—where removing each mechanism raises the admission rate *and* degrades rejection precision. This is a *methods and systems* paper: the contribution is a workflow that any practitioner or researcher can replicate, audit, and extend to a new domain by supplying a different `bottleneck_map.yml`.

2 Related Work

2.1 LLM and Agentic Systems for Investing and Financial Research

Domain-adapted language models have become the baseline for financial NLP. Wu et al. [27] introduced BloombergGPT, establishing strong financial-NLP performance but offering no decision discipline or rejection mechanism [27]. Yang et al. [29] released FinGPT as an open-source counterpart [29]. Moving to autonomous agents, Ding et al. [7] surveyed LLM trading agents, finding that nearly all evaluation is in-sample relative to the model's training cutoff [7]. Xiao et al. [28] built TradingAgents, a bull/bear-debate multi-agent framework evaluated on backtest Sharpe [28]; Zhang et al. [31] contributed FinAgent, a multimodal agent reporting large backtested profit improvements [31]. None impose a pre-registered hypothesis contract, an explicit admission-rate target, or forward validation; their headline metrics are backtest Sharpe ratios rather than the disciplined rejection of poor ideas. Two 2025–2026 results sharpen exactly this point. Ye et al. [30], in *The Alpha Illusion*, argue that reported alpha from end-to-end LLM trading agents should not be treated as deployment evidence until it survives structural-validity tests (temporal integrity, frictions, robustness, calibration, execution), and recommend positioning the LLM as an upstream information interface feeding independent risk/execution modules — not as the decision engine [30]. Lee et al. [16], *Your AI, Not Your View*, give the first quantitative analysis of LLMs' inherent investment biases (large-cap and contrarian preference, and confirmation bias that makes models cling to an initial judgment against counter-evidence) [16]. Both motivate our central design choice: the LLM *sources* hypotheses, but deterministic gates and a forward contract, not the model's own confidence, decide whether anything is admitted.

2.2 LLM Failure Modes Relevant to Research

Three failure families matter here. **Hallucination/confabulation:** models fabricate plausible facts beyond their training distribution, carrying direct fiduciary risk. **Sycophancy:** Malmqvist [21] catalogued how LLMs systematically agree with user-expressed views, generating false confidence — critical when the model is asked to rate a thesis it also generated [21]. **Lookahead leakage:** Gao et al. [10] showed that a positive lookahead-accuracy correlation is direct evidence the model saw the answer in pre-training [10]; Li et al. [18] showed strong backtested LLM-agent returns collapse out-of-sample once leakage is controlled [18]. Benhenda [2] contributes *Look-Ahead-Bench*, a regime-generalization benchmark separating genuine predictive skill from memorized patterns, finding standard LLMs exhibit significant look-ahead bias while point-in-time models do not [2]; Roy and Roy [25], *MemGuard-Alpha*, show in-sample accuracy rises with memorization contamination while out-of-sample accuracy falls, and use cross-model disagreement as a contamination signal [25]. Surveying the field, Kong et al. [15] review 164 studies (2023–25) and find no single bias — lookahead, survivorship, narrative, objective, or cost — is explicitly addressed in more than ∼28% of them, and propose a (retrospective, reviewer-facing) structural-validity checklist [15]. At the factor level, Harvey et al. [11] showed most claimed return factors are likely false discoveries from multiple comparisons, requiring a $t > 3.0$ threshold [11]. Our framework treats these as first-class *design* constraints rather than post-hoc review items: deterministic gates before any expensive-model judgment, bear-case-before-recommendation, and forward (not retrospective) validation.

2.3 Pre-Registration and Research-Integrity Methods

Nosek et al. [23] argued the core value of preregistration is the sharp separation of prediction from postdiction [23]. Casey et al. [5] gave the canonical economics pre-analysis-plan application [5].

In finance, McLean and Pontiff [22] documented that published return predictors decay ∼26% out-of-sample and ∼58% post-publication, much of it data-mining bias [22]; work on pre-analysis plans finds they suppress p-hacking only with a *detailed* plan, not registration alone [3]. Closest to our setting, Hofman et al. [12] adapt clinical-trial pre-registration to predictive modeling, naming three reproducibility failures (overlooked contextual factors, data-dependent decisions, unintentional test-set reuse) and giving a lightweight ML pre-registration template [12]. Our **forward-QPOP** protocol extends this tradition with two elements that template lacks: (i) a domain-specific *bottleneck* schema that makes pre-registration portable across investment themes, and (ii) a *forward* outcome commitment — every hypothesis is content-hashed before the market opens, evaluated on live paper-trading fills rather than historical simulation, and the ledger is append-only — closing the loophole of testing on periods that postdate the model's training cutoff.

2.4 Multi-Agent LLM Orchestration and Verification

Du et al. [8] formalized multiagent debate to improve factuality [8]. Li et al. [17] surveyed "LLM-as-a-judge," identifying self-enhancement bias when generator and judge are the same model [17]. Crucially, more judges do not mean less bias: Ma et al. [20] show debate-based multi-agent judging *amplifies* position, verbosity, chain-of-thought, and bandwagon biases after the first round [20], and Zhao et al. [32] measure sycophancy in *agentic financial* pipelines, finding most models abandon a correct analytical position once a user preference is expressed [32]. These results argue that an adversarial frame must be *structurally enforced*, not left to emergent debate — which is exactly our bear-case-before-recommendation rule. The most methodologically adjacent work is POPPER [13], which uses LLM agents to run sequential *falsification* experiments with Type-I error control [13]. We differ on three axes a reviewer will probe: (i) **forward lock vs. retroactive test** — POPPER validates a hypothesis against *already-existing* data and cannot stop a researcher from running it after seeing the outcome, whereas our hypotheses are content-hashed and dated *before* the out-of-sample window opens; (ii) **discovery vs. validation** — POPPER receives a formed hypothesis, while our workflow begins before the hypothesis exists, sourcing candidate themes through the bottleneck schema; (iii) **domain portability** — our schema carries finance-specific, reusable evidence fields (chokepoint dimensions, dated triggers, source tiers) rather than generic statistical falsification. Our design also keeps the cost discipline: a *cheaper* model gates and a *more expensive* model evaluates only gate-passers, not all candidates.

2.5 Thematic / Supply-Chain / Chokepoint Investing; Factor Crowding

Factor crowding is well-established: Volpati et al. [26] quantified crowding in standard factors [26]; Lou and Polk [19] showed "comomentum" predicts the reversal of crowded momentum [19]; Harvey et al. [11] frame the broader factor-zoo / false-discovery problem [11]. Our bottleneck-graph model formalizes chokepoint identification as a structured schema (supply concentration, demand inelasticity, substitutability, pricing power), and the confidence decomposition tracks crowding and valuation as *gating* dimensions, so a thesis can be rejected on crowding grounds before any simulation.

2.6 Trustworthy / Responsible AI in Finance

Fritz-Morgenthal et al. [9] connected trustworthy-AI principles (fairness, explainability, robustness, accountability) to financial risk management, arguing self-learning models must archive full data and metadata after material changes [9]. Khatchadourian [14] shows empirically that decision determinism and task accuracy are *uncorrelated* in tool-using LLM financial agents and that schema-first architectures give the most audit-compliant consistency [14], while Buscemi et al. [4] map EU AI

Act high-risk requirements to concrete technical-verification activities and document the gap between regulatory intent and current practice [4]. Regulatory direction (EU AI Act 2024; NIST AI RMF) treats high-risk AI as requiring documented, auditable decision trails — a standard backtest-centric agents do not meet. Our content-hashed, timestamped, forward-validated, human-readable ledger makes the system's reasoning inspectable independently of the LLM that produced it.

2.7 Live LLM-Trading Benchmarks and Auditable Trade Logs

A reviewer will rightly note that neither *forward evaluation* nor *tamper-evident logging* is, on its own, novel — so we state precisely where the line is. Qian et al. [24], *When Agents Trade*, introduce the Agent Market Arena, a **lifelong, real-time multi-market benchmark** for LLM trading agents, finding that agent architecture drives risk behavior more than the base LLM [24]. Chen et al. [6], *StockBench*, give a contamination-free sequential trading benchmark and find most LLM agents fail to beat buy-and-hold [6]. On the engineering side, the open-source `llm-quant` system records LLM paper-trades in a **SHA-256 hash-chained, git-tracked ledger** [1]. We borrow from this last design — our Forward-QPOP ledger is likewise hash-chained — but our contribution is *not* the immutable log. The distinction, which we make central: these systems evaluate or record **what an agent *did*** in the market; ours forward-**locks a researcher-level hypothesis** (the binding-chokepoint claim, dated evidence, and entry/update/exit triggers) to a hash-stamped record **before the evaluation window opens**. Live evaluation tells you the outcome; the forward lock makes the *prediction* auditable against that outcome, closing the gap that lets backtest- or benchmark-based claims be rationalized after the fact.

Gap We Fill

Existing work addresses individual validity threats in isolation: lookahead-bias diagnostics and benchmarks [2, 10], structural-validity checklists applied *post-hoc* by reviewers [15], automated falsification for *already-formed* hypotheses [13], output-layer auditability for tool-using agents [14], and live trading benchmarks or hash-chained trade logs that record *executions* rather than *pre-registered predictions* [1, 6, 24]. To our knowledge, no prior system ties these into a *prospective, forward-locked* workflow that unifies all four capabilities: (a) agentic open-ended hypothesis discovery across thematic domains; (b) deterministic gating that enforces source-tier, purity, and overlap rules *before* any expensive model is invoked; (c) **forward pre-registration — not backtesting — as the evidentiary standard**, immutable once content-hashed and dated; and (d) a portable cross-domain bottleneck schema that makes the process reproducible and auditable without relying on any single LLM instance or training snapshot. The combination is what converts the LLM from a decision engine into an auditable sourcing tool. The closest peer, POPPER [13], shares the falsification spirit but validates a static hypothesis against existing data; it has neither the forward temporal lock (which prevents backfitting) nor the discovery phase nor the finance-portable schema.

3 Framework

The framework turns open-ended, LLM-driven discovery into *disciplined, auditable, falsifiable* research by composing five mechanisms. Each exists to neutralize a specific, well-documented failure mode of using LLMs for investment research.

Mechanism	**Failure mode it neutralizes**
Bottleneck graph + confidence decomposition	"right theme → buy the ticker"; theme-chasing
Cheap-source → deterministic-gate → expensive-evaluate funnel	cost blow-up; shallow, ungrounded judgment
Bear-case-before-recommendation	sycophancy; one-sided narratives
Forward-QPOP pre-registration	HARKing; data-snooping; backtest overfitting
No-stories discipline + admission restraint	confabulation; over-admission; narrative drift

3.1 The bottleneck graph (the portable thesis)

A domain is a **graph of physical chokepoints**. Nodes are bottlenecks (a step that is hard to bypass, capacity-constrained, supplier-concentrated); edges encode dependency. The graph is domain-agnostic: AI accelerators today, rare-earth refining / power transformers / LNG / uranium / copper tomorrow — each is *a different `bottleneck_map.yml`*, not a different codebase.

Each node carries: a physical rationale, `bottleneck_dims` (`physical_indispensability`, `substitutability`$^{-1}$, `capacity_lead_time`, `supplier_concentration`, `pricing_power`), a `demand_score`, node-level `valuation_adjustment` / `crowding_adjustment`, and **measurable `exit_triggers`, each with a checkable** `data_source` (an admission rejects any trigger you cannot actually check).

What makes a chokepoint an *edge*: uncertainty about the constraint

A binding chokepoint is necessary but **not sufficient** for a tradeable edge. The edge lives in the **uncertainty about the constraint** — how much of the scarcity is *not yet known or priced.* The question is not only *"is this a chokepoint?"* but *"how anticipated is it?"*

- **Physical chokepoints** (refining capacity, fab lead-times, permits, ore grades) keep their pricing power because their magnitude and timing are **uncertain and discovered late** — the market learns the bottleneck is binding only as lead-times stretch and inventories empty. The `capacity_lead_time` and `supplier_concentration` dims are proxies for exactly this *unpriced* scarcity.
- A **synthetic / scheduled constraint** with a *known date and a known, shrinking magnitude* gets **arbitraged toward zero edge** — the market prices it in earlier each time it recurs. The canonical case is Bitcoin's *halving*: a real algorithmic supply cut, but the most pre-announced supply event in finance, whose marginal shock shrinks each cycle (≈1.7% → 0.83% → 0.41% of supply) *and* which the market now front-runs (a pre-halving all-time-high in 2024). Real constraint, no durable edge.

This is the same principle as **policy-as-evidence**: an event *already in the price* is sunk information, not alpha (a signed award that has already fired, an extended/anticipated cycle). It is why the `valuation_adjustment` and `crowding_adjustment` exist — they discount a constraint the market has already learned. **Prefer chokepoints whose binding is still being discovered** (lead-times stretching, the re-rate ahead) over ones whose schedule everyone already trades.

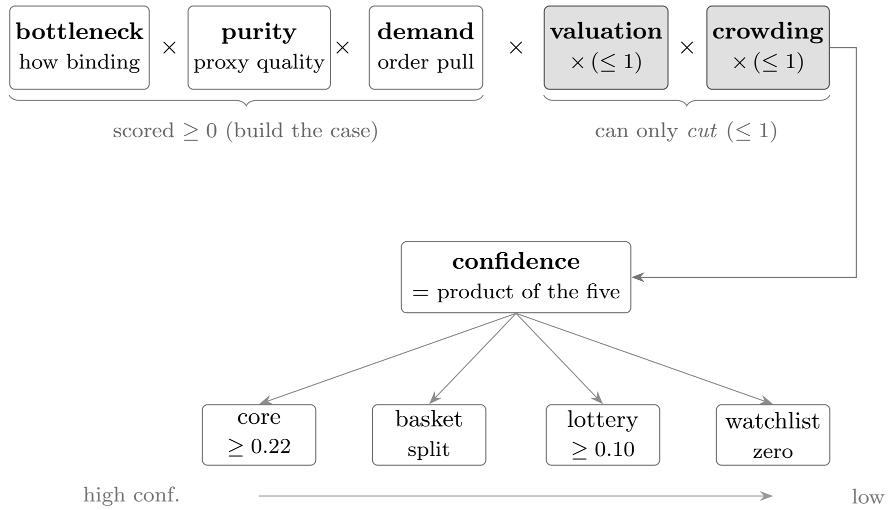

Figure 1: The confidence decomposition. Three evidence-scored factors and two multiplicative penalties capped at 1 (a crowded or richly-valued name can only lose confidence, never gain it) combine into a single score, from which the position tier — and thus sizing (core ≥ 0.22, satellite ≥ 0.10) — is *derived*, not asserted.

3.2 Confidence decomposition (kills theme-chasing)

$$\text{confidence} = \underbrace{\text{bottleneck\_score}}_{\text{node: how binding}} \times \underbrace{\text{exposure\_purity}}_{\text{ticker: captures economics}} \times \underbrace{\text{demand\_score}}_{\text{node: order visibility}}$$
$$\times \underbrace{\text{valuation\_adjustment}}_{\leq 1\text{: penalize priced-in}} \times \underbrace{\text{crowding\_adjustment}}_{\leq 1\text{: penalize crowded}}$$

A correct node with a poor proxy (low purity) or a crowded entry yields *low* confidence. Tier is **derived** from confidence (e.g. core ≥ 0.22, satellite ≥ 0.10, below $\rightarrow$ dropped), not asserted (Figure 1).

Policy is *evidence*, not a factor

Government policy (subsidies, awards, export controls, strategic-autonomy programs) is a real, sometimes durable driver — but it enters as **dated evidence + a monitor trigger**, never as a multiplicative `policy_adjustment`. A multiplier manufactures a number from a *forecast* (is it durable? bipartisan?) — the no-stories failure mode in a lab coat. Two cases instead:

- **Tailwind on a name with a real chokepoint** $\rightarrow$ record a (loader-inert) `policy:` evidence block + a wired `policy_reversal` exit trigger (clawback / milestone miss / reversal). It does **not** resize the name.
- **Underdog whose thesis *is* policy** $\rightarrow$ a *separate, capped* sleeve, admitted via the normal gates, never core.

Fact vs announcement: a *signed, dated, material* award is citable evidence; a political headline / pledge is watchlist-context only. The clinching lesson: a multiplier *rewards the existence* of policy — but a signed award that has *already fired* (in the price) with its clawback removed is sunk information turned into a *liability*; the evidence-and-trigger framing forces the right questions (*has it already fired? what would reverse it? does it unlock capacity?*).

4 Forward-QPOP Protocol

QPOP is a forward hypothesis ledger. It adapts pre-registration and registered-reports discipline from empirical science into agent-assisted investment research, *without* relying on a backtest.

Why forward, not backtest

For a young or structurally-shifting universe (recent IPOs, a thesis the market is actively re-rating), a long backtest is fragile: regime change, survivorship, and lookahead leakage make it easy to overfit a story. Pre-registration replaces "fit the past" with "commit, then observe the future." The ledger is the contract against **HARKing** (hypothesizing after results are known) and **data-snooping**.

For an *LLM-scored* engine the objection is sharper than fragility: a backward test is **structurally invalid**, not merely noisy — and three properties defeat it, each sufficient on its own. **(i) No information is injected backward.** The engine is a deterministic composition of LLM judgments, so retrodicting historical names measures the model's *hindsight*, not the engine's edge. **(ii) The labels are already in the model.** The outcomes one would test against have settled and are present in the model's training corpus, so there is no clean train/test split: the label leaks into the feature through pretraining. **(iii) The features *are* the fingerprint.** The very attributes the engine scores — which chokepoint, what purity, whose capacity — uniquely identify the company and hence its known outcome, so anonymizing the inputs cannot restore a blind test. The consequence is not "prefer forward"; it is that *no* backward procedure can evidence the engine's predictive signal — a backward run is at best an internal-consistency check. This is the engine-level analogue of the HARKing argument the ledger already enforces at the hypothesis level, and it is why we operationalize forward evaluation with the registry below rather than a historical analog study.

The forward-scoring registry (engine-level evaluation)

The hypothesis ledger validates individual *admissions* forward. To evaluate the *scoring engine itself* — whether a higher `final_confidence` ranks a higher forward return — we keep a separate **append-only forward-scoring registry**. At a cohort date we freeze every scored name's effective tier (core/satellite, assigned by the *production* tiering rule on post-refresh values, not a bare threshold) together with its decomposed confidence, and then let outcomes settle *after* the model's knowledge. Because the signatures are recorded forward and the outcomes are unanticipated, the test is honest in exactly the way a backward retrodiction (§ above) is not. The registry is **provisional** until it accrues power — on the order of dozens of name–horizon observations spanning both high- and low-confidence names — so it is a multi-year instrument, reported with the structural-validity checklist, never a near-term gate. A subtle requirement: the cohort must be labeled with the same tiering function the live book uses (including any hold/override path), or the registry's own labels disagree with the book it claims to validate.

Entry types

1. **Admission** — a node or ticker enters the map (especially at meaningful weight) *only* via a pre-registered hypothesis: the binding-chokepoint claim, dated evidence and source tier, the expected return driver, the decomposed confidence, and **measurable exit triggers committed in advance**. No pre-registration → no capital.

2. **Belief update** — every confidence change is a recorded, **content-hash-protected** posterior update with cited evidence. *Tertiary sources alone cannot move confidence.*

3. **Outcome** — when a trigger fires or the thesis ages into consensus, the entry is closed **Supported / Weakened / Falsified**, with the observed metrics. **Closed entries are immutable — open a new id to revise.**

Entry schema (illustrative)

```
id: H-<DOMAIN>-<NN>
created: <date>
status: open            # open | supported | weakened | falsified
role_admitted: satellite   # never straight to core
claim: "<the binding-chokepoint claim, in one sentence>"
mechanism: "<why it is hard to bypass>"
decomposed_confidence:
  bottleneck_score: 0.0
  exposure_purity: 0.0
  demand_score: 0.0
  valuation_adjustment: 0.0
  crowding_adjustment: 0.0
  final_confidence: 0.0
objective_gates:
  { tradable: true, min_price: true,
    min_dollar_volume: true, overlap_penalty: 0.0 }
exit_triggers:
  - { id: supply_easing,
      metric: "...", op: "<",
      data_source: { name: "...", tier: secondary } }
  - { id: demand_slowing,
      metric: "...", op: cut,
      data_source: { name: "...", tier: primary } }
  - { id: crowding,
      metric: "price vs estimate-revision gap", op: ">",
      data_source: { tier: market_implied } }
prior_confidence: 0.0
evidence:
  - { summary: "...", tier: primary, date: <date>, url: "..." }
content_hash: "<hash of the frozen fields>"
prev_hash:    "<content_hash of the previous ledger entry>"
entry_hash:   "<sha256(content_hash + prev_hash)>"
```

The hash chain

The load-bearing fields (`claim`, `exit_triggers`, `decomposed_confidence`, `prior`) are hashed at creation (`content_hash`). Each entry **also binds the previous entry's hash** (`prev_hash`), so the ledger is a **hash chain**:

$$\texttt{entry\_hash} = \mathrm{sha256}(\texttt{content\_hash} \,\|\, \texttt{prev\_hash})$$

not merely a set of independently-hashed rows. Two properties follow:

- **Per-entry integrity:** any later edit to a *closed* entry breaks its `content_hash` and is rejected — an external reviewer can verify the hypothesis was registered *before* the outcome.
- **Append-order integrity:** because each entry commits to its predecessor, you cannot silently insert, delete, or re-order a past entry without breaking every `entry_hash` after it. This is the difference from a bare "immutable trade log" (e.g. a hash-chained *execution* ledger): here the chained object is the *pre-registered hypothesis contract* — claim, dated evidence, and

entry/update/exit triggers committed before the evaluation window — so the chain proves *what was predicted, and when*, not just what was traded.

Design note: the hash-chain construction is borrowed from tamper-evident LLM trade logs; the contribution here is chaining the *hypothesis* (with its forward triggers), not the trade.

Dated admission, update, and exit triggers

Each trigger committed at admission has three mandatory fields:

Field	**Type**	**Purpose**
`id`	string	unique label within the entry
`metric`	string	the observable quantity checked against the threshold
`op`	string	comparison operator or named rule (e.g. `<`, `cut`)
`data_source.name`	string	named source for the metric
`data_source.tier`	enum	`primary` | `secondary` | `market_implied`

Admission trigger (entry): hypothesis registered before any capital deployed.
Update trigger (belief update): posterior revision with cited evidence; tertiary sources alone blocked.
Exit trigger: pre-committed rule; outcome recorded as Supported/Weakened/Falsified.

Forward validation and posterior recording

After each forward observation window:

- Did a pre-registered trigger fire? → reduce/exit per the rule, record the outcome. *Never a story.*
- Did the thesis age into consensus (valuation/crowding adjustments collapse)? → **Weakened**.
- Did the chokepoint hold and the driver play out? → **Supported**.

The accumulated (hypothesis → decision → forward outcome) tuples form a reusable, auditable **dataset** — both the empirical core of the methods paper and the fuel for calibrating the gates on realized outcomes (the self-improvement loop).

5 System Implementation

The framework's five mechanisms are realized as a staged, model-tiered *cascade* — cheap, deterministic filters first, the expensive model last.

The Source→Gate→Triage→Evaluate→Adjudicate Cascade

The discovery pipeline is a five-stage funnel in which each tier hands the next a smaller, higher-value candidate set:

`SOURCE`	→	`GATE`	→	`TRIAGE`	→	`EVALUATE`	→	`ADJUDICATE`
cheap LLM		no LLM		cheapest LLM		mid LLM		expensive LLM
wide, deduped		deterministic		drop low-purity		bear-case-first		admit-flags only

- **SOURCE** (cheap model): high-volume, low-judgment candidate cards with dated, source-tiered evidence — one lens per node, plus gap / social / policy / **literature** lenses. **Deduped against a persistent "seen" set** (every previously-held / watchlisted / rejected name): the lenses skip a decided name unless it shows a *material, dated* change, so the expensive sourcing tokens are spent on genuinely new candidates, not re-researching the book.
- **GATE** (no LLM): tradeability + liquidity + **overlap penalty** vs. the existing book. Pure, reproducible, cheap. A name failing the gate is watchlist-only, never "free capacity."
- **TRIAGE** (cheapest LLM): a quick purity pre-filter that drops the obvious non-starters (diversified-giant-tiny-segment / commodity price-taker / pre-revenue hype) *before* the expensive bear-case pass. **Safe by asymmetry** — a wrong drop only watchlists a name (a delay), and a wrong pursue costs one cheap evaluation that watchlists it anyway.
- **EVALUATE** (mid model, *triage survivors only*): deep fit + a **bear case written before the recommendation** + a falsifiable decision.
- **ADJUDICATE** (expensive model, *admit-flags only*): a capital-at-risk re-judgment of the names the EVALUATE pass flags to admit/replace. Reserve the scarce model for exactly the irreversible call.

Model allocation is a tunable cascade: each tier hands the next a smaller, higher-value set, so the cheapest models do the volume and the expensive model only ever judges admits. The asymmetry is safe throughout — a false-watchlist merely delays a name, while a false-admit is caught downstream.

Confidence Decomposition

A correct bottleneck node with a poor proxy ticker (low exposure purity) or a crowded entry yields *low* confidence. The tier assigned to a name is **derived** from the confidence score (e.g. core ≥ 0.22, satellite ≥ 0.10, below $\rightarrow$ dropped), not asserted by the analyst. The decomposition is:

$$\text{confidence} = \underbrace{\text{bottleneck\_score}}_{\text{node: how binding}} \times \underbrace{\text{exposure\_purity}}_{\text{ticker: captures economics}} \times \underbrace{\text{demand\_score}}_{\text{node: order visibility}} \times \underbrace{\text{valuation\_adjustment}}_{\leq 1\text{: penalize priced-in}} \times \underbrace{\text{crowding\_adjustment}}_{\leq 1\text{: penalize crowded}}$$

No-Stories Discipline and Forward Validation

- **Source tiers:** primary (filings/transcripts) > secondary (reputable trade press) > market-implied (price/volume) > tertiary (social — *idea seed only*, never establishes confidence). A confidence move requires $\geq$ secondary or market-implied corroboration.
- **Overlap penalty:** every candidate is judged *against the book you already hold* — a great theme that duplicates a held bet must **replace, not stack** (gate = comparison, not veto).
- **Structural vs. cyclical:** distinguish a hard chokepoint (few suppliers, multi-year capex) from a commodity-cycle peak riding a bullwhip; the latter is the trap that *looks* like the winner.
- **A position changes only on** a fired pre-registered trigger, a binding cap, or a risk limit — **never a narrative.**

- **Validation is forward:** track-record metrics are judged on *conservative shadow fills* (opening print + assumed slippage), benchmark-relative, with the accumulated ledger as the dataset. The system is optimized for *auditability*, not cleverness — every weight must be explainable by score × purity × adjustments × caps × drift, or no position is taken.

Why the Admission Rate Is the Headline — and Why It Is Not Enough Alone

A naïve LLM screener over-admits by construction — it is rewarded for *finding* ideas. This framework is rewarded for *refusing* them: most candidates are on-theme yet fail "does this improve the book after overlap, purity, valuation, and crowding?" The low admission rate, achieved by pre-committed gates rather than after-the-fact judgment, is one measurable contribution.

But a low admission rate is **not, by itself, evidence of quality** — a system can reject 95% of ideas by being arbitrarily conservative. So restraint is reported with two companions:

- **Rejection quality (rejection precision).** A held-out auditor (an independent LLM and/or a human), shown only the *bull* case and the decision — **not** the engine's bear case — labels each sampled rejection JUSTIFIED or a FALSE REJECTION, with a category (low-purity / duplicate-overlap / valuation-or-crowding / commodity-cycle / pre-revenue-or-hype / unverifiable-or-untradeable). The headline pairs *admission rate* (how much is rejected) with *rejection precision* (whether it should have been). High restraint + low rejection precision would be a failure, not a success.
- **Ablation against baselines.** The same candidate stream is run through an ungated screener, debate-only, no-Forward-QPOP-lock, and no-overlap-penalty variants; the claim is that the full pipeline admits *fewer* ideas **and** that its rejections are better justified — not merely that gates reduce the count.

Outcome decisions over many triggers and positions use a **sequential test with explicit Type-I control** (an anytime-valid / e-value formulation), so monitoring many exit triggers does not inflate false "Falsified" calls — the prospective analogue of the factor-zoo multiple-comparisons problem.

6 Experiment Plan (Pre-Registered)

This plan is itself pre-registered: it commits the evaluation **before** the forward data is collected, consistent with the framework's own discipline. Edit only by appending dated amendments.

Hypotheses

- **H1 (restraint):** the gated, forward-registered pipeline admits a **low fraction** of the candidates it *evaluates* (and a lower fraction of those it *sources*). Pre-committed threshold: admission rate among evaluated finalists $<$ **10%** over the study window. (The prose "low" and the numeric threshold are deliberately the same single-digit target — no looser fallback.)
- **H2 (each discipline contributes):** ablating any one of {deterministic gate, bear-case-first, overlap penalty} **raises** the admission rate vs the full pipeline. *Supported for an ablation if* its admission rate exceeds the full pipeline's by $\geq$ **25% relative OR** $\geq$ **5 percentage points, whichever is larger** (the max guards against a tiny relative jump on a tiny base, and against a large relative jump that is still trivial in absolute terms).

- **H3 (portability):** the disciplines transfer to a second domain with only `bottleneck_map.yml` changed. *Supported if* (a) the second domain's admission rate stays **within 2**$\times$ the first domain's rate, **and** (b) $\geq$ **70%** of the second domain's rejections map to the **same gate / rejection-category taxonomy** used in the first (tradeability, liquidity, low purity, overlap, valuation, crowding, unverifiable-trigger); **and** (c) the second-domain run uses the *same engine code and prompt templates*, with changes limited to `bottleneck_map.yml`, the domain-specific source lists, and the benchmark config — otherwise "portability" could be satisfied by quietly modifying the system. "Comparable" is thus defined on rate, the failure-mode taxonomy, and a no-source-change constraint, not asserted qualitatively.

- **H4 (forward, reported honestly):** over the forward window, benchmark-relative excess return and information ratio on conservative shadow fills are reported with confidence intervals. *No threshold is pre-set for H4 with a small sample* — it is descriptive, not a success gate.

- **H5 (rejection *quality*, not just quantity):** restraint is only evidence of discipline if the rejected ideas were actually bad. We therefore audit rejections: an independent reviewer (a held-out LLM auditor and/or a human), given only the *bull* case and the decision — **not** the engine's bear case — labels each sampled rejection JUSTIFIED or a FALSE REJECTION and assigns a category (low-purity / duplicate-overlap / valuation-or-crowding / commodity-cycle / pre-revenue-or-hype / unverifiable-or-untradeable). *Supported if* **rejection precision** $\geq$ **0.80** ($\geq$ 80% of sampled rejections judged justified) **and** the false-rejection rate is $\leq$ **10%**. This directly answers "a system can reject 95% of ideas by being arbitrarily conservative" — H1 measures *how much* is rejected, H5 measures *whether it should have been.*

Design

- **Domains:** the AI-supply-chain worked example, then 1–2 additional bottleneck domains as configs.

- **Horizon:** $\geq$ 6 months of forward observation per domain before any performance claim; discipline metrics reported from the first runs.

- **Unit of analysis:** a candidate (sourced $\rightarrow$ gated $\rightarrow$ evaluated $\rightarrow$ admitted/not) and a position (admitted $\rightarrow$ forward outcome).

- **Primary metrics:** sourced/gated/evaluated/admitted counts; admission rate; no-action rate; source-tier distribution; overlap-penalty distribution; ledger-integrity (% admissions with a valid hash, chained to the prior entry, registered before the position); **rejection precision** and **false-rejection rate** from the H5 audit.

- **Secondary (forward):** excess return vs the theme benchmark; information ratio; active drawdown; gross/net thematic beta; turnover; shadow-fill slippage.

- **Ablations and baselines:** rerun the same candidate stream against (a) **ungated LLM screener** (no deterministic gate); (b) **debate-only** (bull/bear with no pre-registration lock); (c) **no Forward-QPOP lock** (same evaluation, hypotheses not content-hashed before the window); (d) **no overlap penalty**; and, where feasible, (e) a **human/manual thematic shortlist** for the same domain. Report each one's admission rate **and** its rejection precision — the point is to show the full pipeline admits *fewer* ideas *and* that its rejections are better justified, not merely that gates reduce count.

Decision Rules (Pre-Committed)

- H1 is **supported** if the evaluated-finalist admission rate is $<$ **10%** over the window.
- H2 is **supported** (per ablation) if that ablation's admission rate exceeds the full pipeline's by $\geq$ **25% relative or $\geq$ 5 percentage points, whichever is larger**.
- H3 is **supported** if the second-domain admission rate is **within 2**$\times$ the first-domain rate, **and** $\geq$ **70%** of its rejections map to the shared rejection-category taxonomy, **and** the second-domain run uses the *same engine code and prompt templates*, with changes limited to `bottleneck_map.yml`, the domain-specific source lists, and the benchmark config (clause (c) — otherwise "portability" could be satisfied by quietly modifying the system).
- H5 is **supported** if **rejection precision** $\geq$ **0.80** and the **false-rejection rate** $\leq$ **10%** on the audited sample.
- A position's outcome is recorded **Supported** / **Weakened** / **Falsified** strictly by its pre-registered exit triggers — never by an after-the-fact narrative. Because a thesis is judged against *several* triggers over time, outcome decisions use a **sequential test with explicit Type-I error control** (an e-value / anytime-valid formulation, [13]), so that monitoring many triggers across many positions does not inflate false "Falsified" calls — the multiple-comparisons analogue of the factor-zoo problem, controlled *prospectively*.

Implementation Details (Fixed in Advance, for Reproducibility)

Recorded here so the forward log is reproducible and leakage-resistant; any change is a dated, additive amendment.

- **Model versions & prompts:** the exact model identifier (and version/date) for each tier (SOURCE / first-pass EVALUATE / adjudication) is logged per run; prompt templates are committed in the repo and referenced by commit hash. A model-version change is an amendment, not a silent swap.
- **Source timestamps:** every evidence item carries the publication date of the source and the retrieval date; an admission may rely only on evidence dated **on or before** the admission date.
- **Price source:** a single declared price vendor (e.g. daily adjusted closes) with corporate actions (splits/dividends/spin-offs) applied via the vendor's adjusted series; the vendor is named in the run config.
- **Paper-trading timestamp & fill rule:** admissions take effect at the **next session's opening print** after the content hash is registered (never the same bar as the signal). Forward track-record metrics use a **conservative shadow fill** = opening print + **assumed slippage (default 10 bps)**; broker paper fills validate automation only and are not the metric of record.
- **Transaction cost / slippage:** the 10 bps slippage assumption plus declared commission (default \$0) are applied to every shadow fill; turnover is reported so costs are auditable.
- **Corporate actions:** handled via the adjusted price series; admissions/exits triggered by a corporate action (e.g. acquisition close) are logged as such, not as thesis outcomes.
- **Post-admission news:** the system **may** read post-admission news, but it may only *fire a pre-registered exit trigger or open a new dated belief-update entry* — it may **not** retroactively edit the original hypothesis or its triggers (the content hash makes such an edit detectable).

Integrity Controls

- Every admission carries a hash-chained QPOP entry (each entry binds the prior entry's hash) registered before the position — tampering with any past entry breaks the chain.
- The ledger is append-only; closed entries are immutable.
- No metric is changed after seeing results; amendments are dated and additive.
- Forward results are **evaluated** on conservative shadow fills, not optimistic fills.
- **Structural-validity reporting [30]:** any forward performance number is reported only alongside its structural-validity checklist — temporal integrity (admission strictly precedes the evaluation bar), frictions (slippage + costs applied), robustness (across the accumulated ledger, not a cherry-picked window), calibration (admitted confidence vs realized outcome), and execution (shadow-fill rule). A return without this checklist is not reported as evidence.

Threats to Validity (Declared in Advance)

Small sample / wide intervals early; LLM nondeterminism across model versions; survivorship in which domains get built; the admission-rate metric is *necessary but not sufficient* for alpha. The paper reports these rather than hiding them.

7 Worked Example: AI Supply Chain

We instantiate the framework on the AI-accelerator supply chain. One discovery round, end to end: roughly forty candidate cards are sourced across twelve lenses; the persistent seen-set leaves about ten genuinely new symbols; the deterministic gate passes about six (foreign-only listings and sub-liquidity names are rejected mechanically); a cheap triage drops the obvious low-purity names; and the bear-case-first evaluation plus adjudication admit *none*. The modal outcome is no action. Across the pilot build-out (including earlier sessions, not only this 0-admit round), we show three representative dispositions: **admitted** — a supplier-concentrated critical-material chokepoint, sized at half its unconstrained satellite weight by a separate concentration cap (restraint expressed as sizing); **rejected** — an integrated energy major proposed for a "helium chokepoint" whose helium revenue is under one percent of the company (a real chokepoint the *ticker* does not capture); and **overlap-penalized** — a high-purity optical pure-play routed to a replace-not-stack decision because its node sleeve already held the incumbent it would duplicate. A full sanitized walk-through is the released `examples/ai_supply_chain/` domain.

8 Results

> **Pilot metrics — not a performance claim, and not confirmatory.** This paper validates *process discipline*, not investment profitability; investment validity requires the H4 forward window. The numbers below are **pilot / initial discipline metrics** from early candidate batches: they *motivate* the registered evaluation and are labeled as such, *not* treated as confirmatory evidence. Where a threshold (H1's `<10%`, H5's $\geq$`0.80`) was not locked strictly before these specific batches were observed, the batch number is a pilot reading against that threshold, and the pre-registered plan governs only the forward study window. Forward returns are not yet observed.

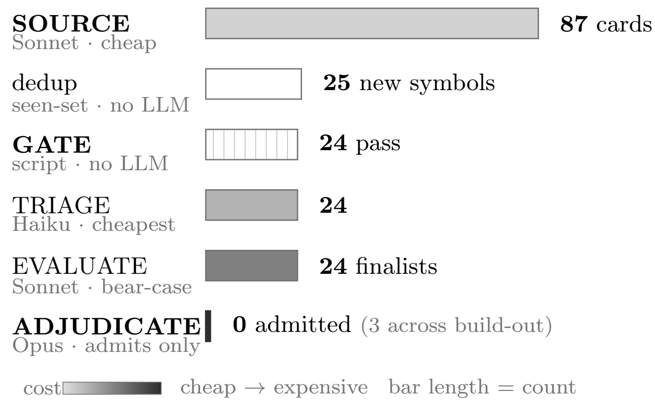

Figure 2: The discovery cascade on a real batch (2026-06-07, AI-supply-chain). Bar *length* is the candidate count surviving each stage; bar *fill* is the model cost tier (white = deterministic, no LLM). The dedup step removes duplicates, not ideas; the expensive model judges only the survivors and admitted **none** this session — this is rejection, not throughput, and "no action" is the modal outcome.

8.1 The Funnel, on a Real Batch

A discovery session (2026-06-07, AI-supply-chain domain) over under-covered chokepoint layers, sourced across 12 lenses (per-node, gaps, social, policy, and literature), produced the cascade in Figure 2: ∼87 candidate cards collapsed to ∼25 unique new symbols after seen-set dedup; 24 passed the deterministic gate (tradeability + liquidity + overlap; foreign-OTC and sub-threshold-liquidity names rejected mechanically); and the bear-case-written-before-recommendation evaluation plus adjudication admitted **0** this session (prior build-out sessions admitted 3, all at small satellite).

Admission rate among evaluated finalists **this session: 0/24** (rule-of-three 95% upper bound ≈ 12% on so small a sample); trailing admission rate across the build-out is **low single digits** — a pilot reading consistent with H1 (`< 10%`), to be confirmed over the forward window with proper intervals. Two consecutive 0-admit rounds were *not* a failure of sourcing; the gate-passers were diversified mega-caps (chokepoint = a tiny segment) or pre-revenue, and the genuinely pure vehicles surfaced were foreign-OTC and failed the US-tradeability gate. "No action" is the modal outcome.

8.2 Rejection-Quality Audit (H5)

The central objection to a low admission rate is that *a system can reject 95% of ideas by being arbitrarily conservative.* So a **held-out, bull-only LLM auditor** (a different model instance — an *adversarial diagnostic*, not a claim of model-independent audit; human review is planned for the forward study, not yet done) was shown only the steelmanned **bull** case and the decision — **not** the engine's bear case — and asked to label each sampled rejection JUSTIFIED or FALSE REJECTION with a category.

The reported metric, in order of precedence, is: **primary** — raw auditor precision 0.93 (Wilson 95% CI $[0.69, 0.99]$); **secondary** — post-adjudication resolution 1.00, reported only to show the layering works; **interpretation** — the single auditor/engine disagreement is a useful diagnostic signal, not a number to celebrate. The takeaway is the raw 0.93, *not* an "effective 14/14."

Category distribution of justified rejections: low-purity-conglomerate ×8, commodity-cycle ×2, pre-revenue/hype ×2, duplicate-overlap ×1.

The one flagged false rejection is reported, not hidden — it is the most useful output.

Metric	Value
Sampled rejections audited	14
Rejection precision (raw, primary)	**0.93** (13/14; Wilson 95% CI $[0.69, 0.99]$)
Disagreement rate	0.07 (1/14)
Adjudicated precision (secondary)	1.00 (14/14)

Table 1: Rejection-quality audit summary.

The auditor judged an early-stage pure-play (a primary-sourced design win, no conglomerate dilution) admissible at small satellite where the engine had said "watchlist," and noted a **mild systematic over-caution on early-stage pure-plays.** That single disagreement is a credibility signal — the audit is adversarial, not a rubber stamp.

We treat the raw 0.93 as the primary metric and do not lean on the 1.00. Because the audit's verdict is a *flag*, not a final decision, the flagged name went to a capital-at-risk **adjudication** (the expensive model, fresh web verification), which surfaced the fact the bull-only auditor had missed — the name's headline design win is **shared across a 14-member supplier alliance that already includes a held name**, so it is one of fourteen substitutable suppliers, not a chokepoint owner — and **upheld the rejection.** A process that overrules its own auditor and then reports 100% would read as circular, so we headline the **raw auditor precision (0.93)** and report the adjudicated 1.00 only as a *secondary* resolution mechanism. The 0.93 is already good and does not need the 1.00. The methodological point is the architecture, not the number: a cheap, bull-biased audit *should* occasionally push to admit, and the expensive layer exists to catch that — the same cheap-screen → expensive-adjudicate asymmetry that governs admissions.

8.3 Ablation: Each Discipline Contributes to Restraint (H2)

The same **38-candidate batch** was run through baselines that each *remove one discipline*, and their admission rates compared to the full pipeline. To keep the comparison honest: the candidate cards, the model identifier, and the temperature are *identical* across arms; every arm may choose ADMIT / WATCHLIST / REJECT under the same capital-at-risk context — only the decision *discipline* differs. All arm prompts are released (`src/prompts.md`) so the baselines can be inspected; they are explicitly *allowed to reject*, not strawmen forced to admit.

Arm	Admitted	Rate
Full pipeline (gate + seen-set + triage + bear-case-first + overlap)	**0 / 38**	**0%**
− overlap penalty (judge each name on its own, ignore duplication with the held book)	25 / 38	66%
− bear-case-first (recommend from the bull thesis, no written bear case)	28 / 38	74%
Ungated LLM screener (same cards + admit/watchlist/reject, but no gate, purity bar, overlap penalty, or bear-case requirement)	**38 / 38**	**100%**

Table 2: Ablation results on the 38-candidate batch.

Reproducibility of the baselines. The ungated baseline was **allowed to reject** — it is not a strawman forced to admit. The full baseline prompt, the model version, the temperature, and the candidate cards are released (`src/prompts.md`; cards sanitized) so a reader can judge for themselves whether the ungated arm was unfairly admission-biased rather than taking our word that the 100% is a fair comparison.

The result is monotone and decisive: in this batch and prompt setting, the **ungated screener admitted every candidate** (the "over-admits by construction" failure made concrete); removing *any single* discipline raises the admission rate far past H2's threshold ($\geq$ 25% relative *or* $\geq$ 5 pp); and the full stack drives 100% $\rightarrow$ 0%. This is the comparison the headline needs — the pipeline admits **fewer** ideas, and the raw rejection audit (0.93) shows the ones it keeps out are mostly the right ones. H2 is **supported** for every ablated discipline (a *pilot* reading; the registered ablation locks thresholds before the forward batches). We read the 100% honestly: an ungated screener that is free to reject but admits everything is exhibiting exactly the structural bias-to-admit the framework exists to counter — the informative quantity is the *change* in behavior when a discipline is removed, not that any single arm always admits.

> A reviewer's exact objection — "a system can reject 95% of ideas by being arbitrarily conservative" — is answered by these two results together: the ablation shows the rejections come from *specific, removable disciplines* (not blanket conservatism), and the audit shows they are *justified*.

8.4 Portability (H3): A Second Domain

To probe H3 we ran the *same* funnel on a deliberately distant domain — **pharmaceutical injectables / GLP-1 drug delivery** — changing only the node list. The funnel narrowed the same way: **27 candidate cards sourced** $\rightarrow$ 17 with a US-tradeable symbol (10 were real chokepoints whose only vehicle is foreign / private, the same "no clean domestic vehicle" failure seen in the AI domain) $\rightarrow$ deterministic gate **7 pass / 10 fail** (foreign-OTC ADRs + sub-liquidity) $\rightarrow$ triage + bear-case + adjudication $\rightarrow$ **0 admit, 2 watchlist, 5 reject**.

The disciplines transferred. The engine *located* the domain's canonical chokepoint on the first pass — a coated-elastomer closure near-monopoly whose components are named in FDA/BLA filings, so substitution requires multi-year change-control revalidation (the pharma analogue of the substrate-film chokepoint in the AI domain) — and scored it high on both bottleneck strength and purity. It **rejected the five conglomerate look-alikes** on *low exposure purity* (the dominant rejection category in both domains; the chokepoint is a small segment of a diversified parent). And the **entry discipline fired independently of the chokepoint-quality test**: the bear-case pass flagged the pure chokepoint for admission, but adjudication *downgraded it to watchlist* on exactly the valuation/crowding logic the first-domain book applied to its own real-but-priced-in names — a stretched multiple with a thesis that reduces to "the multiple holds."

Measured against the pre-registered H3 criteria: **(a)** the second-domain admission rate ($\approx$ 0) is within 2$\times$ the first domain's; **(b) 100%** of the second-domain rejections map to the shared taxonomy (low-purity-conglomerate, foreign / no domestic vehicle, illiquid, priced-in-watchlist). **(c)** is only *partially* met and we report it as such: the disciplines are domain-general, but the current SOURCE / EVALUATE prompt templates embed the first domain's framing, so the probe required authoring domain-general prompts rather than changing `bottleneck_map.yml` alone. True config-only portability requires threading a `domain` parameter through the prompt templates — a concrete, declared next step. The portability evidence is therefore at the *method* level (the disciplines and the rejection taxonomy transfer), with config-level portability still to be demonstrated.

A third domain, and a new rejection pattern. As a further portability probe we ran the same funnel on a third, distant domain — the launch / satellite-internet supply chain, grounded in the recent S-1 of its dominant operator. The engine transferred unmodified and again produced a decisive negative (**0 admit**). Two of the prior domains' patterns recurred and a new one emerged. **(i)** The rejection taxonomy held: low-purity conglomerates, foreign / no-US-vehicle chokepoints (one UK-listed and several Asia-listed), and priced-for-perfection cyclicals trading near all-time highs were all rejected on the same gates. **(ii)** The engine caught a class of error a naive "supplier-of-X" screen does not — it flagged apparent suppliers that are in fact *competitors* or *spectrum counterparties* of the target rather than input vendors. **(iii)** A substitution mode appeared that the prior domains never exhibited: the customer *vertically integrating the chokepoint away.* The target actively engineers out its own named inputs (in-sourcing propellant pressurization, primary structures, and user-terminal silicon), so a "supplier" thesis must additionally survive the customer building the part itself. The engine absorbed this without modification, adding *"can the customer in-source this input?"* to the bear case alongside the second-source, new-technology, and policy-substitution tests. A third distant domain therefore reinforces the *method*-level portability claim — the disciplines and the rejection taxonomy transfer — and the new bypass mode shows the bear case is extensible to domain-specific substitution paths rather than fixed to the first domain's.

A fourth probe, across *substrates*. As a stronger portability test we extended the same disciplines to a *non-physical* substrate — proprietary-data and network monopolies. Because that probe is design-and-backtest stage rather than live, we report it separately in Appendix A: the disciplines transferred (the same purity and over-payment gates separated genuine data monopolies from contestable look-alikes), and it surfaced a substrate-specific refinement to the *entry* rule with a clean negative result.

8.5 What Is Still Pending (Declared)

- **Remaining baselines** (debate-only, no-Forward-QPOP-lock) — the three run here (ungated, bear-case-off, overlap-off) cover the core H2 disciplines; the remaining two are additive.
- **Forward outcomes** (H4) — accrue over the forward window; reported only with the structural-validity checklist.
- **Forward-scoring registry read-out** — the engine-level cohort is seeded and frozen forward (the first cohort is recorded); it is provisional until it accrues sufficient name–horizon observations, with the first read-out roughly two quarters out, and is likewise reported only with the structural-validity checklist.

8.6 Reproducibility and Release

To make "open release" concrete, Table 3 states exactly what is and is not published. The boundary is firm: the *method* (code, prompts, schemas, synthetic and sanitized fixtures) is released so the disciplines and the ablations are reproducible; the *live book* (positions, broker data, paid feeds) is never released, because publishing it would read as "copy these trades," which this project explicitly is not.

Artifact	Released?	Contains
Code	Yes	gate, scoring, ledger, example runner
Prompts	Yes	source / triage / evaluate / adjudicate / audit
Candidate cards	Sanitized	redacted evidence summaries
Ledger	Synthetic	QPOP schema and sample hashes
Live positions	No	not released
Broker data	No	not released
Paid data	No	not redistributed

Table 3: What the open release contains.

9 Discussion

Cross-domain portability. Adding a domain is *intended* to be mostly configuration-driven. The pilot second- and third-domain runs (§Portability) support the *method-level* claim — the same engine located the canonical chokepoint and rejected the conglomerate look-alikes on a different node list, the rejection taxonomy transferred, and a domain-specific bypass mode (customer in-sourcing) was absorbed into the bear case unmodified — but is honest about the limit: *true* config-only portability still requires threading a `domain` parameter through the SOURCE/EVALUATE prompt templates, which currently embed the first domain's framing, so the probe required authoring domain-general prompts rather than changing `bottleneck_map.yml` alone. The disciplines (gates, source tiers, bear-case-first, overlap, the forward lock) are domain-agnostic; a full config-only domain swap (the same engine on rare-earth refining, power transformers, uranium, or copper from a new map alone) is a declared next step, not a demonstrated result.

Uncertainty about the constraint (a conceptual case study). A binding chokepoint is *necessary but not sufficient* for a tradeable edge — the edge lives in the *uncertainty* about the constraint. Physical chokepoints keep pricing power because their magnitude and timing are discovered late; a *synthetic, scheduled* constraint with a known date and a shrinking magnitude is arbitraged toward zero edge. The cleanest illustration is Bitcoin's halving: a real algorithmic supply cut, but the most pre-announced supply event in finance, front-run and with a geometrically decaying marginal shock. This is the same principle as policy-as-evidence — information already in the price is, to that extent, not alpha — and it is why the framework prefers chokepoints whose binding is still being discovered. We present it as an illustration of why *forward, unanticipated* evidence matters, not as a contribution.

Forced-cheap entry (cross-substrate; see appendix). The constraint-uncertainty principle above has a testable, cross-substrate form, which we pursue separately in Appendix A: a point-in-time backtest on proprietary-data businesses indicates the unpriced gap comes from *involuntary forced selling*, not from the market's belated *recognition* of a hidden data business — a generalization probe with a clean negative result, not a performance claim.

10 Limitations

The forward sample is small and early; intervals are wide and no live-money track record is claimed. LLM outputs are nondeterministic across model versions, so we log the exact model identifier and prompt commit per run and treat a version change as a dated amendment. Survivorship enters through which domains get built. Most importantly, the admission rate is *necessary but not sufficient* for alpha: a disciplined rejection process is a precondition for, not a guarantee of, forward performance, which only the pre-registered forward log can establish. Finally, a screening engine's *coverage* (which chokepoints the SOURCE stage surfaces) is a separate failure mode from its *judgment* (whether the gates correctly accept or reject a surfaced candidate). In the third-domain probe a blind comparison against an external expert reading list revealed a genuine chokepoint the SOURCE stage had missed even though the gates rejected it correctly once it was surfaced — a coverage gap, not a judgment gap. We treat that distinction as load-bearing: a coverage gap is closed by adding a source lens, never by loosening the gates, and we use blind held-out comparisons against external lists as the way to detect coverage gaps when entering a new domain.

11 Ethics and Responsible Use

This is a research framework, not investment advice, and we make no performance claim. The system is optimized for auditability: each admitted position is explainable from its content-hashed ledger entry (we release the schema and synthetic/sanitized examples, not the live ledger). We comply with data-provider terms (no paid feeds are redistributed), publish only synthetic and sanitized examples, and never release live positions, broker credentials, or the scored live map. See `ETHICS.md` and `DISCLAIMER.md`.

Conflicts of interest and disclosure. The author maintains a private research book that applies this method and may hold positions in names within the example domains discussed here. Example names are illustrations of the decision taxonomy — not endorsements, recommendations, or a disclosure of the book; no live positions, weights, or sizes are revealed. Where example cases are named (Figure 3), they are historical, closed dislocations presented for their measured valuation facts; current or live candidates are described only generically. The author declares no other competing interests.

12 Conclusion

LLMs make thematic research cheap but unreliable as decision engines. We presented a forward-registered, auditable workflow that converts an LLM from a decision engine into a sourcing tool: deterministic gates, source-tiered dated evidence, a bear case written before any recommendation, a hash-chained pre-registration ledger, and forward (not backtest) validation. The headline is restraint — the agent rejects most of what it surfaces, for reasons an independent auditor judges defensible, and an ablation shows that each discipline contributes to that restraint. The framework, a worked example, and a domain template are released for others to fork and extend.

A Cross-Substrate Portability Probe: Proprietary-Data Monopolies

The domains in § Portability vary the node list but share a *physical* substrate — factories, fill-finish lines, launch hardware. As a stronger portability test, we ran the same disciplines on a *non-physical* substrate: proprietary data and network monopolies, where the chokepoint is a dataset an industry cannot rebuild (registration and title histories, citation graphs, contributor-fed credit files) rather than a plant. This probe is design-and-backtest stage, not live — we pre-registered the gates, then ran a *point-in-time*, public-data-grade backtest over historical, closed dislocations in such businesses. The disciplines transferred: the same purity and over-payment gates separated genuine data monopolies from contestable look-alikes, and rejected celebrated data businesses that were never actually cheap.

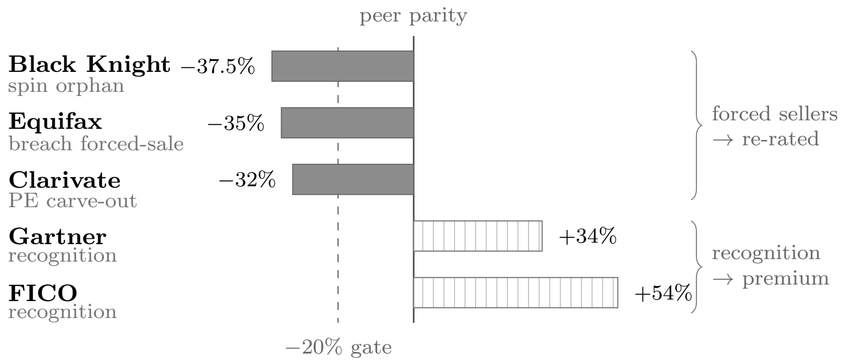

Figure 3: **Forced selling, not recognition** (portability probe, fourth substrate). Valuation gap to information-services peers at the dislocation date (T_0), for historical, closed dislocations in proprietary-data businesses; point-in-time, public-data grade. Forced-seller dislocations (solid) — a spin-off orphaned by index funds, a post-breach forced sale, a private-equity carve-out — left real ∼30–40% discounts that subsequently re-rated; *recognition* dislocations (hatched), where the market belatedly discovered a hidden data business, traded at *premiums* at T_0, leaving no gap. The dashed line is the pre-registered −20% entry gate. Retrospective case studies illustrating the entry mechanism — not assessments of the businesses or forward price views.

Forced-cheap entry, and a cross-substrate negative result. The constraint-uncertainty principle has a testable, cross-substrate form. A chokepoint is mispriced only when its price is *involuntarily* dislocated, and the dislocation mechanism is substrate-specific. Physical chokepoints get cheap on a capital-spending *cycle*; proprietary-data chokepoints compound steadily and rarely cycle cheap, so their only forced-cheap window is an *observable forced seller* — a spin-off orphaned by index funds, an index deletion, a private-equity exit. A point-in-time backtest (public-data grade) sharpens this into a falsifiable claim and yields a negative result that mirrors the halving case in the Discussion (Figure 3): *forced-selling* dislocations left real ∼30–40% valuation gaps to information-services peers that subsequently re-rated, whereas *recognition* dislocations — the market belatedly discovering a hidden data business — traded at *premiums*, leaving no gap to capture. Recognition is an *anticipated* re-pricing, arbitraged just as the pre-announced halving is; only the *involuntary* seller leaves the unpriced gap. The result narrows the mechanism rather than broadening the claim — the same kept-disproof discipline as the rejection-quality audit — and we offer it as a generalization probe, not a performance claim.

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